National Retail Corp (NRC)—Enterprise GenAI Platform Scenario Brief

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Executive Summary

National Retail Corp (NRC) is a Fortune 500 national retailer operating 1,200+ stores across North America, with annual revenue of $48B. Following a successful pilot phase, NRC is consolidating three independently developed AI initiatives into a unified, enterprise-grade GenAIOps platform. This scenario brief establishes the organizational context, application profiles, constraints, and stakeholder priorities that will govern your architecture design.

1. Company Profile

Attribute	Detail
Industry	Multi-channel Retail (Brick-and-Mortar + E-commerce)
Annual Revenue	$48B USD
Store Count	1,247 retail locations + 3 distribution centers
Employees	85,000 (12,000 corporate, 73,000 field)
Digital Footprint	Mobile app (8M MAU), e-commerce site (4.2M monthly visits), in-store kiosks (5,200 units)
Cloud Strategy	Azure-first; multi-subscription landing zone architecture via Azure Landing Zones (ALZ)
Data Platform	Azure Synapse Analytics, Azure Data Lake Gen2, Microsoft Fabric (2026 GA features)
Security Posture	SOC 2 Type II, PCI-DSS Level 1, NIST CSF 2.0 aligned; Zero Trust Network Architecture (ZTNA)
AI Maturity	Level 3 (Defined)—moving toward Level 4 (Quantitatively Managed) per Microsoft AI Maturity Model

2. Business Context and Strategic Drivers

- NRC's Board of Directors approved a $34M AI transformation budget for FY2026–2027. The Chief Digital Officer (CDO) and CFO have jointly mandated that all AI deployments meet enterprise operational standards by Q3 2026. Key drivers include:

- Peak Season Resilience: Black Friday/Cyber Monday (BFCM) drives 40% of annual digital revenue. AI systems must sustain 15× normal traffic without degradation.
- Cost Control: Cloud AI spend reached $2.1M in Q4 2025 without centralized governance. The CFO requires per-application cost allocation, quota enforcement, and showback reporting.
- Compliance and Audit: The General Counsel requires all HR-facing AI to maintain immutable audit trails, data residency (US-only), and GDPR/CCPA-aligned data subject rights.
- Operational Consolidation: Three AI teams (E-commerce, Supply Chain, HR) currently operate siloed pipelines, causing duplicated infrastructure, inconsistent monitoring, and conflicting model versions.

3. Application Profiles

3.1 AskRetail—Customer-Facing E-commerce Assistant

Dimension	Specification
Purpose	Natural language product discovery, order tracking, returns initiation, and personalized recommendations across mobile app and website.
User Base	8M mobile MAU; 45% of e-commerce sessions engage the assistant.
Traffic Pattern	Baseline: 12,000 conversations/day. Peak (BFCM): 180,000 conversations/day. Burst ratio: 15×.
Latency SLA	p50 ≤ 800ms, p99 ≤ 2,500ms (end-to-end, gateway to final response).
Availability SLA	99.95% uptime (21.6 min/month max downtime).
Data Sensitivity	Moderate—processes PII (names, addresses, order history), payment context (last-4 digits only), but no full PCI data.
Model Stack	Azure OpenAI GPT-4o (primary), GPT-4o-mini (fallback for high-traffic queries). Azure AI Search (vector + semantic hybrid) for product catalog.
Risk Classification	TIER 1—Customer-facing revenue-critical. Highest operational scrutiny.
Stakeholder Priority	CDO: zero-downtime deployments. CMO: response quality and brand safety. CFO: cost per conversation must not exceed $0.04.
Compliance	CCPA, PCI-DSS adjacent (no CHD storage), ADA digital accessibility.

3.2 StockSense—Inventory Forecasting Assistant

Dimension	Specification
Purpose	Generates demand forecasts, reorder recommendations, and supply-chain risk summaries for inventory planners and buyers.
User Base	340 internal inventory planners, 45 supply chain analysts.
Traffic Pattern	Batch-heavy: 80% of requests are scheduled morning reports (6:00–8:00 AM ET). Ad-hoc: ~400 queries/day during business hours.
Latency SLA	Batch: completion within 45 minutes. Ad-hoc: p95 ≤ 8 seconds (complex multi-step reasoning acceptable).
Availability SLA	99.5% uptime (3.6 hrs/month max downtime). Batch window must complete by 9:00 AM ET.
Data Sensitivity	Internal Confidential—sales velocity, supplier pricing, margin data. No customer PII.
Model Stack	Azure OpenAI o3-mini (reasoning for complex forecasting), GPT-4o (summary generation). Azure ML managed endpoints for custom time-series models. SQL-based retrieval from Azure Synapse.
Risk Classification	TIER 2—Internal data-intensive. Accuracy prioritized over speed.
Stakeholder Priority	CSCO: forecast accuracy (MAPE < 8%). CFO: inventory carrying cost reduction target $12M/year. CTO: integration with existing SAP ERP.
Compliance	SOX (financial data lineage), internal data classification policy (Level 2).

3.3 HRHelper—Employee Support and Policy Assistant

Dimension	Specification
Purpose	Answers HR policy questions, guides employees through benefits enrollment, leave requests, and escalation paths. Handles sensitive inquiries (harassment reporting, accommodation requests).
User Base	85,000 employees (corporate + field). Peak: open enrollment period (Nov–Dec) drives 3× baseline.
Traffic Pattern	Baseline: 3,200 conversations/day. Peak: 9,500/day. Highly conversational; average 6.2 turns per session.
Latency SLA	p95 ≤ 3,500ms. Users tolerate slightly higher latency for sensitive topics if empathy/accuracy is high.
Availability SLA	99.9% uptime (43.8 min/month max downtime). Must be available 24/7 for global shift workers.
Data Sensitivity	HIGHLY SENSITIVE—SSN, salary, medical information, performance reviews, harassment reports. Subject to attorney-client privilege for legal escalations.
Model Stack	Azure OpenAI GPT-4o with enterprise data grounding via Azure AI Search (HR policy index). Content filters (Prompt Shield, Protected Material Detection) mandatory.
Risk Classification	TIER 1-S—Customer-facing (internal) + Sensitive. Requires strictest governance.

Stakeholder Priority	CHRO: employee trust and confidentiality. General Counsel: litigation hold readiness, audit trail immutability. CISO: zero data leakage, encryption at rest/transit, no model training on employee data.
Compliance	GDPR (EU employees), CCPA, HIPAA-adjacent (wellness program data), EEOC, SOX (compensation equity analytics), NIST AI RMF 1.0.

4. Stakeholder Matrix and Governance Structure

Stakeholder	Role	Primary Concern	Decision Authority
Chief Digital Officer (CDO)	Executive Sponsor	Revenue protection, customer experience, innovation velocity	Final approval on Tier 1 app architecture changes
Chief Financial Officer (CFO)	Budget Owner	Cost per inference, infrastructure ROI, showback accuracy	Approval on >$50K/month infrastructure changes
Chief Information Security Officer (CISO)	Risk Owner	Data protection, threat surface, compliance posture	Veto on security architecture; approves encryption standards
Chief Human Resources Officer (CHRO)	Business Owner (HRHelper)	Employee trust, confidentiality, accessibility	Approval on HRHelper model updates and data handling
Chief Supply Chain Officer (CSCO)	Business Owner (StockSense)	Forecast accuracy, ERP integration, planner productivity	Approval on StockSense data pipelines and model refresh
VP of Platform Engineering	Implementation Lead	Reliability, scalability, operational efficiency	Technical design authority; approves pipeline and orchestrator choices
AI Ethics and Compliance Committee	Governance Body	Fairness, transparency, regulatory alignment	Mandatory review for all production model promotions

5. Enterprise Constraints and Non-Functional Requirements

1. The following constraints are non-negotiable and must be reflected in all architecture designs:
2. Azure-First Policy: All infrastructure must deploy to Azure US East, US West 2, or US Central regions. No multi-cloud except Azure-fronted CDN.

3. Landing Zone Compliance: All resources must deploy into pre-provisioned ALZ management groups with mandatory tagging (CostCenter, Environment, DataClassification, AppName).
4. Identity Standard: Microsoft Entra ID (Azure AD) is the sole identity provider. Conditional Access policies require MFA for production changes. Managed identities preferred over service principals.
5. Network Isolation: Tier 1-S (HRHelper) must run on private endpoints/VNet injection. Tier 1 (AskRetail) may use public endpoints with WAF and DDoS Protection Standard. Tier 2 (StockSense) uses service endpoints.
6. Key Management: Azure Key Vault with HSM-backed keys (Premium SKU) for all secrets, keys, and certificates. Key rotation every 90 days.
7. Backup and DR: RPO ≤ 1 hour, RTO ≤ 4 hours for Tier 1; RPO ≤ 4 hours, RTO ≤ 8 hours for Tier 2. Cross-region replication to paired region.
8. Cost Allocation: Every resource must be tag-compliant for cost showback. Budget alerts at 80%, 100%, and 120% of monthly forecast.
9. Model Governance: All production models must be registered in Azure AI Foundry Model Catalog with version pinning. No "latest" tag references in production.

6. Program Success Metrics (KPIs)

KPI Category	Metric	Target (Q3 2026)
Operational	Mean Time to Recovery (MTTR)	< 15 minutes for Tier 1
Operational	Deployment Frequency	≥ 2 production releases/week per app
Operational	Change Failure Rate	< 5% (rolled back or hotfixed deployments)
Quality	Groundedness Score (AskRetail)	≥ 0.92 (Azure AI Evaluation SDK)
Quality	Hallucination Rate (AskRetail)	< 1.5% of responses
Quality	Forecast MAPE (StockSense)	< 8%
Quality	HRHelper Escalation Accuracy	≥ 95% correct routing to live HRBP
Cost	Cost per Conversation (AskRetail)	≤ $0.04
Cost	Monthly AI Infrastructure Spend	$1.4M (down from $2.1M baseline)

Compliance	Audit Findings	Zero critical findings; ≤ 3 minor findings

7. How to Use This Brief

Use this brief as the authoritative source for all design decisions in your LC 3 project. When selecting rollout strategies, evaluation gates, or observability thresholds, reference the specific SLAs, risk classifications, and stakeholder priorities defined here. A strong submission will demonstrate traceability: every architectural choice should map back to a business requirement, constraint, or risk profile in this document.

References: Microsoft Azure Well-Architected Framework (2026), Microsoft AI Maturity Model, NIST AI RMF 1.0, Azure Landing Zones Documentation, Azure AI Foundry Best Practices.